Sanjit Dandapanthula

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Education

Carnegie Mellon University (CMU) — Pittsburgh, PA
Ph.D. in Statistics and Machine Learning (advised by Aaditya Ramdas and Nicholas Boffi) — *Aug. 2024 – present*

University of California, Los Angeles (UCLA) — Los Angeles, CA
M.A. in Applied Mathematics — *Apr. 2023 – Mar. 2024*
B.S. in Mathematics w/ highest honors, minor in Data Science and Engineering — *Sep. 2020 – Mar. 2023*

Research interests

I am very broadly interested in the theoretical foundations of modern machine learning and AI. Recently, I have been working on problems related to *generative AI*, *deep learning theory*, *optimal transport*, *changepoint detection*, *conformal prediction*, and *multiple testing*.

Experience

Abridge AI — New York City, NY
Research Intern — *May 2026 – present*

- **Active fine-tuning:** Developing new online label collection algorithms for active LLM fine-tuning

Amazon Web Services (AWS) – LogAnalytics and AI Operations — San Jose, CA
Applied Science Intern — *May 2025 – August 2025*

- **Representational similarity of LLMs:** Proposed a novel and theoretically principled method to compare the learned representations of LLMs for knowledge distillation tasks
- **IGW distance between Gaussians:** Proved new formulae for the inner product Gromov-Wasserstein (IGW) distance between Gaussians over infinite-dimensional spaces using optimal transport theory and convex analysis
- **Multimarginal optimal transport:** Created fast and memory-lean algorithms to solve multimarginal optimal transport problems between Gaussians using the Burer-Monteiro factorization, with provable guarantees
- **User clustering:** Applied theoretical results to cluster users using data drawn from real-world text datasets

Jet Propulsion Laboratory (JPL) at NASA — Pasadena, CA (remote)
Graduate Student Researcher (with Mikael Kuusela and Maggie Johnson) — *Jan. 2025 – Jan. 2026*

- **Remote sensing:** Built novel statistical methods to better estimate land surface temperature (LST) using remotely sensed observations from the ECOSTRESS probe, for applications in agriculture and urban planning
- **Computer vision:** Used a vision transformer (ViT) to find boundaries of agricultural fields and segment images
- **Spatiotemporal statistics:** Used parallel computing to solve Gaussian process equations involving images collected over 7 years, yielding high-resolution point estimates and uncertainty quantification of LST

Narya.ai (The Mumble App) — San Jose, CA
Lead Machine Learning Engineer — *May 2024 – Aug. 2024*

- **AI note-taking:** Managed entire ML pipeline for new AI note-taking app in a fast-paced startup environment ($4M seed raised)
- **State-of-the-art LLMs:** Fine-tuned latest LLM models (GPT-4o, Llama 3.1 405B, Claude 3.5 Sonnet) for efficient voice-based editing, re-writing, keyword finding, and search indexing on user notes
- **Transcription:** Used Google's WebRTC-VAD model to reduce transcription hallucination during silence in audio
- **Vision RAG:** Wrote retrieval-augmented generation model to find link to original social media post or article given a user screenshot, or do other helpful contextual research using real-time information from the Internet
- **LLM privacy:** Ported many fine-tuned open-source models to Apple CoreML to use private on-device LLMs

Institute for Pure and Applied Mathematics (IPAM) — Los Angeles, CA
AI Research Intern + Project Manager — *Jun. 2023 – Sep. 2023*

- **Particle methods for AI:** Led a team of 4 researchers to train a network using a particle-type Monte Carlo tree search to solve the cart-pole problem in 10 seconds (roughly 30x faster than a state-of-the-art deep Q-network)
- **High-performance computing:** Wrote low-level systems code in Rust to implement multi-threading and used the CUDA platform to enable fast GPU-level concurrency

David Harold Blackwell Summer Research Institute (DHBSRI) Berkeley, CA
ML Research Intern (with Jelani Nelson – UC Berkeley EECS) *Jun. 2022 – Sep. 2022*

- **Algorithm development:** Developed new algorithms to solve the learning-augmented sorting problem in Levenshtein distance, ∞-normed, and 2-normed metric spaces and wrote a paper proving their optimality

Polymath REU remote
Research Intern (with Zoran Šunić) *Jun. 2021 – Sep. 2021*

- **Computational graph theory:** Led a 10-person team's programming efforts to conjecture and prove recurrence relations describing the spectra of Schreier graphs of self-similar groups, which I solved explicitly

Papers + preprints

Are we really tilting? The mechanics of reward guidance in flow and diffusion models (2026) [arXiv] [code]
SD and N. M. Boffi *submitted, NeurIPS*

Downscaling land surface temperature data using edge detection and block-diagonal Gaussian process regression (2025) [arXiv] [code]
SD, M. Johnson, M. Pascolini-Campbell, G. Hulley, and M. Kuusela *Environmental Data Science*

Gradient descent for deep equilibrium single-index models (2025) [arXiv] [code]
SD and A. Ramdas *in preparation*

Optimal transportation and alignment between Gaussian measures (2025) [arXiv] [code]
SD, A. Podkopaev, S. P. Kasiviswanathan, A. Ramdas, and Z. Goldfeld *submitted, MathOR*

Offline changepoint localization using a matrix of conformal p-values (2025) [arXiv] [code]
SD and A. Ramdas *TMLR*

Anytime-valid FDR control with the stopped e-BH procedure (2025) [arXiv]
H. Wang, SD, and A. Ramdas *Statistics and Probability Letters*

Multiple testing in multi-stream sequential change detection (2025) [arXiv] [code]
SD and A. Ramdas *submitted, Bernoulli*

Talks

Multiple testing in multi-stream sequential change detection. IWSM (2025)
Are we really tilting? The mechanics of reward guidance in flow and diffusion models. Abridge AI (2026)
Downscaling LST using edge detection and block-diagonal GP regression. Climate Informatics (2026)
Gradient descent for deep equilibrium single-index models. LemanTh (2026)
Multiple testing in multi-stream sequential change detection. International Seminar on Selective Inference (2025)
Gromov-Wasserstein geometry of Gaussian measures. AWS LogAnalytics and AI Operations (2025)
Conformal changepoint localization. AWS LogAnalytics and AI Operations (2025)
Multiple testing in multi-stream change detection. 3rd Workshop on Sequential Anytime-Valid Inference (2025)
Multiple testing in multi-stream sequential change detection. AWS LogAnalytics and AI Operations (2025)
Particle-type methods for intelligent game-playing. Joint Mathematics Meetings (2023)
Learning-augmented sorting algorithms. Simons Institute for the Theory of Computing (2022)
Schreier graphs of self-similar groups. Joint Mathematics Meetings (2021)

Teaching

Carnegie Mellon University — Pittsburgh, PA
TA for 36-410 (stochastic processes and Markov chains) — *Jan. 2025 – May 2025*
TA for 36-700 (grad. probability theory and statistics) — *Aug. 2024 – Dec. 2024*

Thomas M. Liggett Advanced Topics Seminar — Pittsburgh, PA
Lead instructor for semester-long graduate seminar on optimal transport (see my notes here) — *Jan. 2025 – May 2025*

Olga Radko Endowed Math Circle (ORMC) — Los Angeles, CA
Lead Instructor for Advanced 1A section — *Sep. 2021 – Mar. 2024*

Extracurriculars + awards

Extracurriculars: Crisis Text Line volunteer, Carnatic flutist, marathon runner, long-distance biker, mountaineer
Awards: UCLA Departmental Scholar (2023), Putnam Exam Top 500 Scorer (2021), National Merit Scholar (2020)